
Notes for ‘The Second Derivative Test’

Important Ideas and Useful Facts:

- (i) **The Second Derivative Test:** Recall that if f and f' are differentiable then we call f' the *first derivative* and f'' the *second derivative* of f .

Suppose the function is well-behaved near $x = c$, by which we mean that the first and second derivatives exist at $x = c$ (and, in fact, the existence of the first derivative implies that f is continuous at $x = c$, which is explained in more advanced courses in calculus).

We have a simple test in two parts:

- (a) If $f'(c) = 0$ and $f''(c) > 0$ then $f(c)$ is a local minimum.
(b) If $f'(c) = 0$ and $f''(c) < 0$ then $f(c)$ is a local maximum.

This follows from the sign diagram for the second derivative, for if $f''(c)$ is positive then the curve is concave upwards, so there will be an apex of the curve corresponding to a minimum, and if $f''(x)$ is negative then the curve is concave downwards, so there will be an apex of the curve corresponding to a maximum.

- (ii) **The Second Derivative Test may not help:** The Second Derivative Test does not allow you to draw conclusions about the existence of a local extremum if $f'(c) = f''(c) = 0$.

For example, the functions f , g and h defined by $f(x) = x^4$, $g(x) = -x^4$ and $h(x) = x^3$ have the following properties:

$$f'(x) = 4x^3, f''(x) = 12x^2, g'(x) = -4x^3, g''(x) = -12x^2, h'(x) = 3x^2, h''(x) = 6x,$$

so that

$$f'(0) = f''(0) = g'(0) = g''(0) = h'(0) = h''(0) = 0.$$

Yet, clearly, $f(0) = 0$ is a (global) minimum for f , $g(0) = 0$ is a (global) maximum for g and $h(0) = 0$ is neither a local maximum nor minimum for h .

Examples:

1. Consider the function f whose rule is the quadratic

$$f(x) = ax^2 + bx + c,$$

where a , b , and c are constants such that $a \neq 0$. Then $f'(x) = 2ax + b$ and $f''(x) = 2a$. The turning point occurs when $f'(x) = 0$, that is, when $x = \frac{-b}{2a}$.

If $a > 0$ then $f''(x) = 2a > 0$, for all x and, in particular, when $x = \frac{-b}{2a}$. In this case, part (a) of the Second Derivative correctly predicts that the turning point corresponds to a minimum, coinciding with the fact that the associated parabola is facing upwards.

On the other hand, if $a < 0$ then $f''(x) = 2a < 0$ for all x and, in particular, when $x = \frac{-b}{2a}$. In this case, part (b) of the Second Derivative Test correctly predicts that the turning point corresponds to a maximum, coinciding with the fact that the associated parabola is facing downwards.

2. Consider the functions f and g whose rules are

$$f(x) = xe^x \quad \text{and} \quad g(x) = xe^{-x}.$$

From the Product Rule, it follows that

$$f'(x) = e^x + xe^x = (1+x)e^x \quad \text{and} \quad f''(x) = e^x + (1+x)e^x = (2+x)e^x.$$

Then $f'(x) = 0$ if and only if $1+x = 0$ (since e^x is always positive), that is, $x = -1$. Observe that

$$f''(-1) = (2-1)e^{-1} = e^{-1} = \frac{1}{e} > 0.$$

By part (a) of the Second Derivative Test, $f(-1) = -e^{-1} = -\frac{1}{e}$ must be a local minimum. In fact, we know more: this is a *global* minimum. One can see this from the sign diagram for $f'(x)$, noting that its sign coincides with the sign of the factor $(1+x)$:

x			-1	
$f'(x)$		$-$	0	$+$



From the Product Rule, and the fact that $\frac{d}{dx}(e^{-x}) = -e^{-x}$, we see that

$$g'(x) = e^{-x} - xe^{-x} = (1-x)e^{-x} \quad \text{and} \quad g''(x) = -e^{-x} - (1-x)e^{-x} = (x-2)e^{-x}.$$

Then $g'(x) = 0$ if and only if $1-x = 0$ (since e^{-x} is always positive), that is, $x = 1$. Observe that

$$g''(1) = (1-2)e^{-1} = -e^{-1} = -\frac{1}{e} < 0.$$

By part (b) of the Second Derivative Test, $g(1) = e^{-1} = \frac{1}{e}$ must be a local maximum. In fact, we know more: this is a *global* maximum. One can see this from the sign diagram for $g'(x)$, noting that its sign coincides with the sign of the factor $(1-x)$:

x			1	
$g'(x)$		$+$	0	$-$



Note that if we had looked at the sign diagrams for the first derivatives, we would have known immediately about the global minimum for f and the global maximum for g , so the Second Derivative Test becomes superfluous. However, the Second Derivative Test is valuable as an extra check or as part of an exploration. Mathematicians are always on the look-out for alternative ways of reaching any given conclusion, to help safeguard against errors or for discovering errors.